

CSE 311 DATABASE SYSTEMS

Introduction

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Instructor Profile

Education:

PhD (Computer Science), University of Waterloo, Canada (ongoing)

MSc (Computer Science), Dalhousie University, Canada

BSc (Computer Science & Electronic Engineering), IUT, Bangladesh

Area of Research: Human Computer Interaction, User-value centric design, Artificial Intelligence, IoT

Reference Book

Fundamentals of Database Systems 7th edition, 2015

Authors: Ramez Elmasri, Shamkant B. Navathe

Pearson ISBN-13: 978-0133970777

Marks Distribution

Assessment Tools	Weightage (%)
Attendance	5
Quizzes	15
Midterm	20
Final Exam	35
Lab Work	10
Term Project	15
Total	100

Course Outcomes

Sl.	CO Description
1	explain conceptual model through entity, relationship diagrams (E-R diagram) and extended ER diagram.
2	use relational algebra in simple and complex queries based on set theory
3	construct code in SQL DDL (Data Definition Language) and , SQL DML (Data Manipulation Language) for table creation and query processing.
4	build a Web-based relational database system, using scripting languages (e.g., PHP) and an open-source database development system (e.g., MySQL).

Your Learning Goal

Go to the following website -

www.menti.com

Use Code: 3335 4436

CHAPTER 1

Databases and Database Users

OUTLINE

- Basic Definitions
- Types of Databases and Database Applications
- Typical DBMS Functionality
- Example of a Database (UNIVERSITY)
- Main Characteristics of the Database Approach

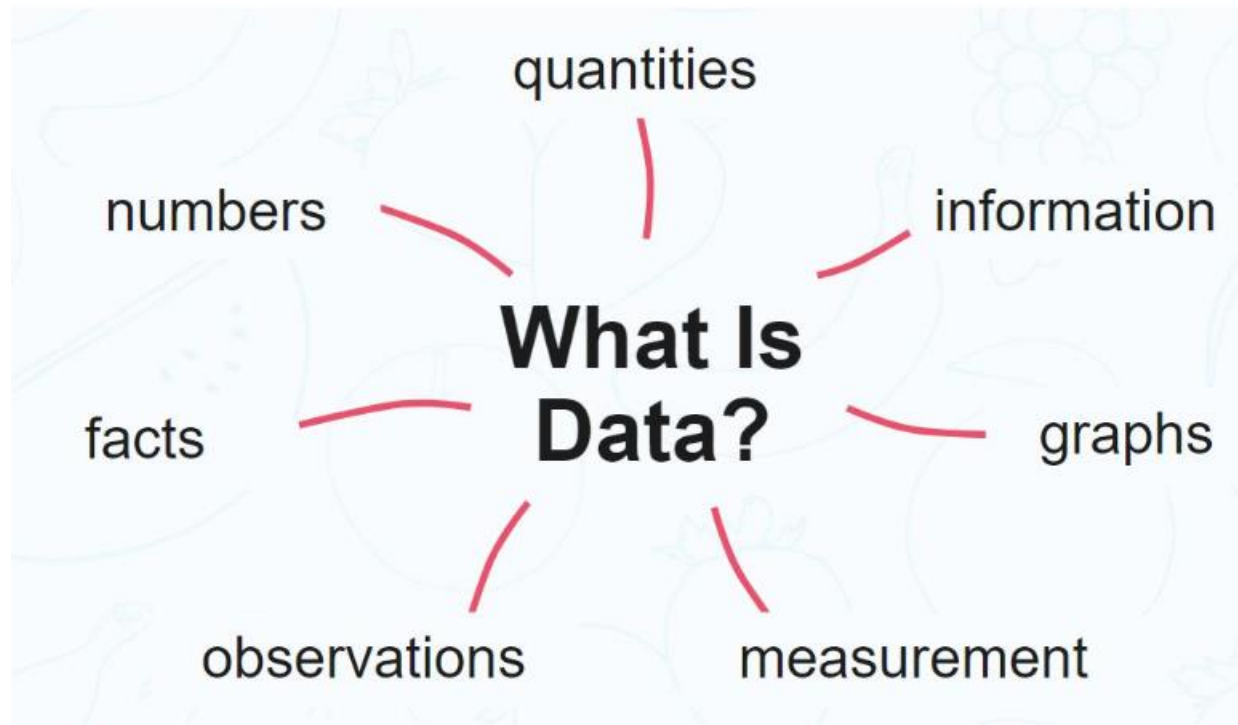
Basic Definitions

- **Database:**
 - A collection of related data.
- **Data:**
 - Known facts that can be recorded and have an implicit meaning.

**What Is
Data?**

Basic Definitions

- **Database:**
 - A collection of related data.
- **Data:**
 - Known facts that can be recorded and have an implicit meaning.



Basic Definitions

- **Mini-world:**
 - Some part of the real world about which data is stored in a database. For example, student grades and transcripts at a university.
- **Database Management System (DBMS):**
 - A software package/system to facilitate the creation and maintenance of a computerized database.
- **Database System:**
 - DBMS software + the data itself + sometimes, the applications

Types of Databases and Applications

- **Numeric and Textual Databases**
 - numerical data, reviews, opinions, views
- **Multimedia Databases**
 - graphics (sketches, drawings), images, video, audio
- **Geographic Information Systems (GIS)**
 - connects data to a map, integrating location data
- **Biological and Genome Databases**
 - protein sequencing, molecular structure, DNA sequences etc.
- **Data Warehouses**
 - enterprise system that analyze data from multiple sources
- **Mobile databases**
 - connected to a mobile computing device

Recent Developments (1)

■ Social Networks

- Facebook (Meta) – posts, photos, videos, reels
- Twitter (X) – tweets
- Instagram - photos
- Linked-In - posts

■ Search Engines

- Google, Bing, Yahoo
 - collect their own repository of web pages for searching purposes

Recent Developments (2)

- New Technologies
 - **Big Data** storage systems involving large clusters of distributed computers
 - transaction processing systems, customer databases, emails, medical records
 - **NOSQL** (Non-SQL, Not Only SQL) systems
 - JSON documents or key-value pairs
 - **Cloud** - huge data centers using thousands of machines.

What is “big data”?

- Big Data
 - properties: volume, velocity, variety
 - insight discovery and process optimization
- Bottom line:
 - Any data that exceeds our current capability of processing can be regarded as “big”

Why is “big data” a “big deal”?

■ Government

- emergency services

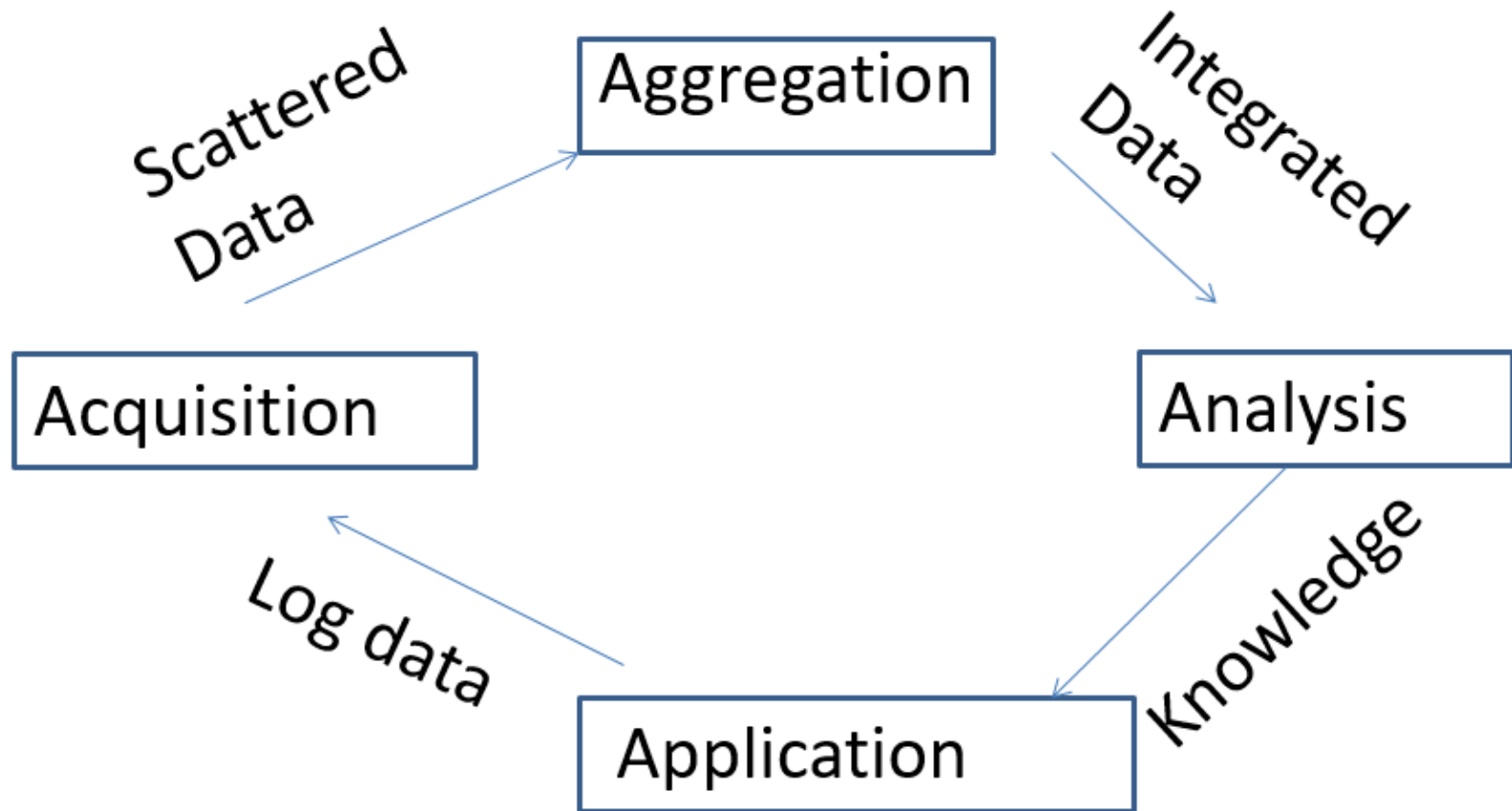
■ Private Sector

- Shopping malls ~1 million customer transactions / hour
- Facebook handles 40 billion photos from its user

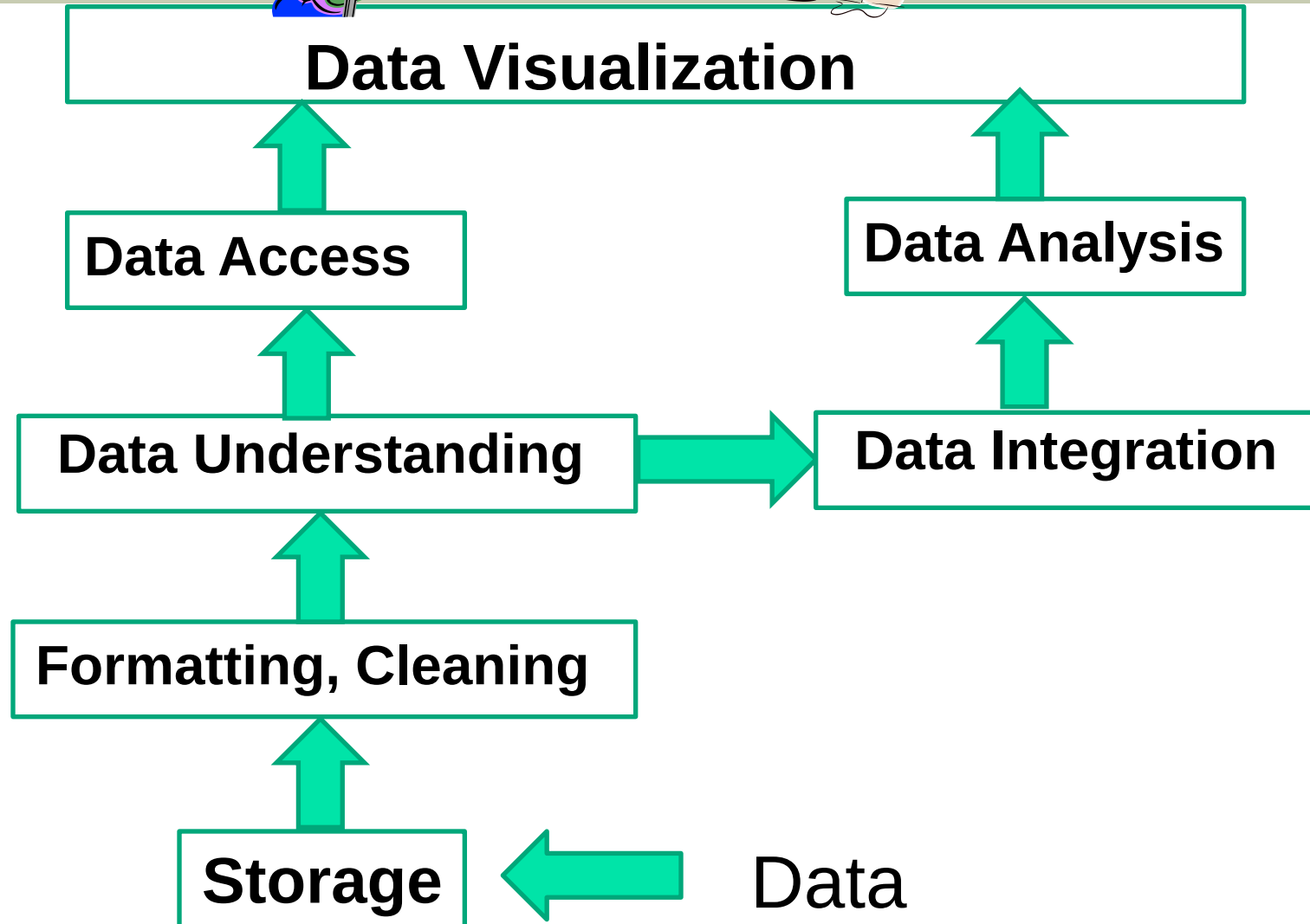
■ Science

- Large Synoptic Survey Telescope will generate 140 Terabyte of data every 5 days
- Biomedical computation like decoding human Genome and personalized medicine

Lifecycle of Data: 4 “A”s



Computational View of Big Data



Big Data & Related Disciplines



Human-Computer Interaction

Data Visualization

Databases

Information Retrieval

Data Access

Machine Learning

Data Analysis

Data Mining

Computer Vision

Speech Recognition

Data Understanding

Natural Language Processing

Data Integration

Data Warehousing

Formatting, Cleaning

Signal Processing

Storage

Information Theory

Many Applications!

Data

Simplified database system environment

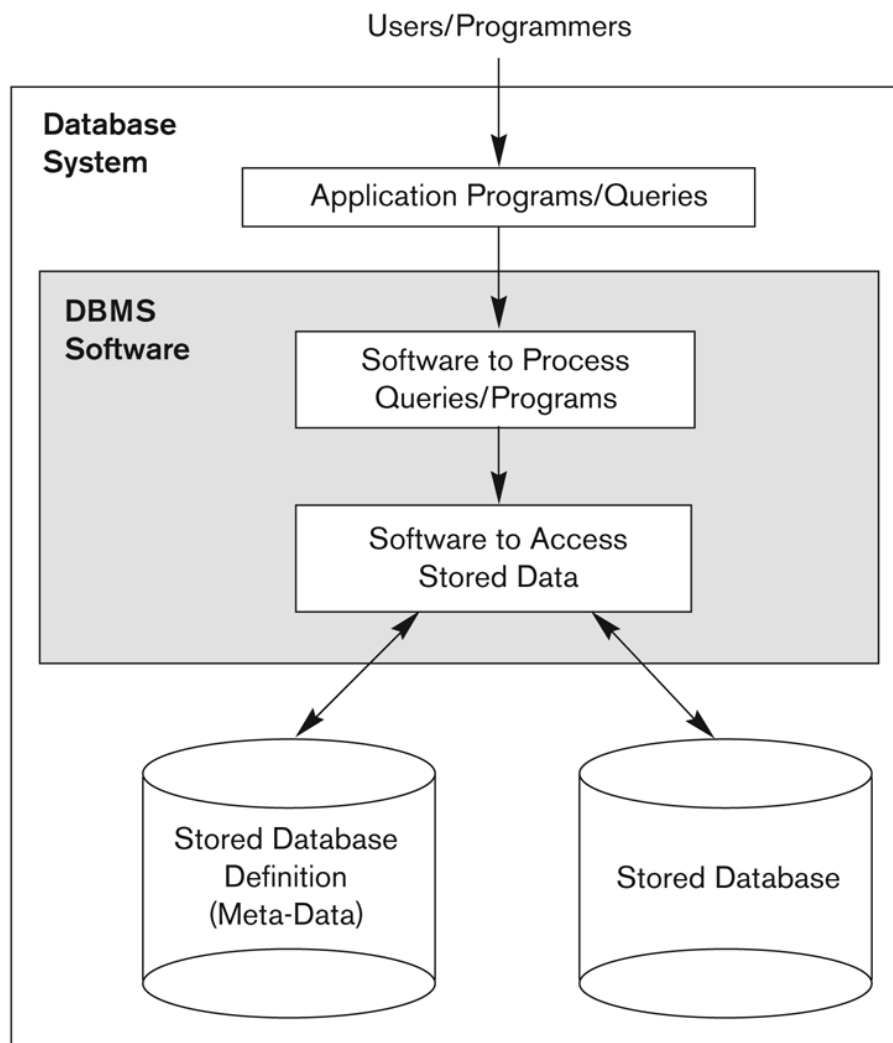


Figure 1.1
A simplified database
system environment.

What does a DBMS do?

Our goal in CSE 311 Lab -

- *Define* a database - data types, structures, and constraints
- *Manipulating* the database
 - Accessing the database through Web applications
 - Retrieval: Querying, generating reports
 - Modification: Insertions, deletions and updates to its content
- Keeping all data valid and consistent

Other DBMS Functionalities

- DBMS may additionally provide:
 - Protection or Security measures
 - Presentation and visualization of data
 - Maintenance of the database

Example of a Database (with a Conceptual Data Model)

- **Mini-world for the example:**
 - Part of a UNIVERSITY environment
- **Some mini-world *entities*:**
 - STUDENT
 - COURSE
 - SECTION (of COURSEs)
 - PREREQUISITE

Example of a Database (with a Conceptual Data Model)

- **Some mini-world *relationships*:**
 - **STUDENTs** *take* SECTIONs
 - **SECTIONs** *are of specific* COURSEs
 - **COURSEs** *have* **PREREQUISITE** COURSEs
 - STUDENTs *have* **GRADE_REPORT**

Example of a Simple Database (University)

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Example of a Simplified Database Catalog

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

Figure 1.3

An example of a database catalog for the database in Figure 1.2.

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

Note: Major_type is defined as an enumerated type with all known majors. XXXXNNNN is used to define a type with four alpha characters followed by four digits

Main Characteristics of the Database Approach

- **Data Abstraction:**
 - hide storage details and present conceptual view
- **Support of multiple views of the data:**
 - Different view for different user interest

Main Characteristics of the Database Approach (continued)

- **Sharing of data and multi-user transaction processing:**
 - Allowing a set of **concurrent users** to retrieve from and to update the database.
 - **Concurrency control** within the DBMS guarantees that each transaction is correctly executed or aborted
 - **Recovery subsystem** ensures each completed transaction has its effect permanently recorded in the database
 - **OLTP** (Online Transaction Processing) is a major part of database applications. This allows hundreds of concurrent transactions to execute per second.

Conclusion

- Basic Definitions
- Types of Databases and Database Applications
- Typical DBMS Functionality
- Example of a Database (UNIVERSITY)
- Main Characteristics of the Database Approach